LAB EXPERIMENT 7

Experiment 7: Part-of-Speech Tagging Using NLTK

Aim

To perform part-of-speech tagging on a sentence using the NLTK library.

Algorithm

1. Start the program.
2. Import NLTK.
3. Download the required tokenizer and POS tagger data.
4. Define a sample sentence.
5. Tokenize the sentence into words and punctuation.
6. Use nltk.pos_tag() to assign a part-of-speech tag to each token.
7. Display the tokens with their tags and stop the program.

Code

```
import nltk

nltk.download("punkt", quiet=True)
nltk.download("punkt_tab", quiet=True)
nltk.download("averaged_perceptron_tagger_eng", quiet=True)

text = "The cat runs quickly."
words = nltk.word_tokenize(text)
tags = nltk.pos_tag(words)

print("Word -> POS Tag")
for word, tag in tags:
    print(word, "->", tag)
```

Output / Screenshot

```
import nltk

nltk.download('punkt')
nltk.download('averaged_perceptron_tagger_eng')
nltk.download('punkt_tab')

text = "The cat runs quickly."

words = nltk.word_tokenize(text)
tags = nltk.pos_tag(words)

print("POS Tags:")
print(tags)
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt.zip.
[nltk_data] Downloading package averaged_perceptron_tagger_eng to
[nltk_data]     /root/nltk_data...
[nltk_data]   Unzipping taggers/averaged_perceptron_tagger_eng.zip.
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt_tab.zip.
POS Tags:
[('The', 'DT'), ('cat', 'NN'), ('runs', 'VBZ'), ('quickly', 'RB'), ('.', '.')]
```

Result

Thus, the NLTK POS tagger was used to assign grammatical tags to the words and punctuation in the given sentence.